Grace Tharp

15 Multiple Integrals

15.1 Double Integrals Over Rectangles

Prerequisite information for this section:

Consider the function

$$f(x) = x - 2\ln x \qquad \text{on} \qquad 1 \le x \le 5$$

Estimate the area between the x-axis and the graph of f using a Riemann sum with **four** approximating rectangles ($n = 4$), and taking the sample points to be the **right endpoints**. In calculus 1, you may have referred to this estimate as R_4. (You may want to review your calculus 1 notes and section 5.1 of your text.)

$f(2) = 2 - 2\ln(2)$

$f(3) = 3 - 2\ln(3)$

$f(4) = 4 - 2\ln(4)$

$f(5) = 5 - 2\ln(5)$

$\Delta x = 1$

$f(2) + f(3) + f(4) + f(5) = 2+3+4+5-2(\ln(2)+\ln(3)+\ln(4)+\ln(5))$

$= 14 - 2\ln(120) \approx 4.425$

Suggest a way to extend the method of Riemann sums to two-dimensional domains.

Evaluate a function on a grid of (x,y) values and change Δx to $\Delta x \Delta y$

Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

- **Integrable (function of two variables):** [Discussion above Box # 6 in Section 15.1; be sure to **write out the limit** that must exist.]

If the limit $\lim_{m,n \to \infty} \sum_{i=1}^{m} \sum_{j=1}^{n} f(x_{ij}^*, y_{ij}^*) \Delta A$ exists, then f is integrable.

All continuous functions are integrable, and discontinuous functions may also be integrable sometimes.

- **Midpoint Rule for double integrals:** [Box before Example 3 in Section 15.1.]

$$\iint_R f(x,y)\,dA \approx \sum_{i=1}^{m}\sum_{j=1}^{n} f(\bar{x}_i, \bar{y}_j)\,\Delta A$$

where $\bar{x}_i$ is the midpoint of $[x_{i-1}, x_i]$

- **Partial integration:** [Paragraph before Box # 7 in Section 15.1.]

For $\int_a^b f(x,y)\,dx$, we hold y fixed and integrate wrt x only.

- **Iterated Integral:** [Use the discussion before, within, and after Boxes # 8 and # 9 in Section 15.1.]

integrating multiple times:

$$\int_a^b \int_c^d f(x,y)\,dx\,dy$$

- **Average Value of a (two-variable) function:** [Discussion above Figure 17 in Section 15.1.]

$$f_{avg} = \frac{1}{A(R)}\iint_R f(x,y)\,dA$$

where $A(R)$ is the area of the region R.

Topics to learn in this section:

1. The definition and properties of the double integral.
2. Volume interpretations of double integrals.
3. Fubini's Theorem.
4. Average Value of a function of two variable.

15.2 Double Integrals Over General Regions

Prerequisite information for this section:

There is no formal warm-up for section 15.2. However, this is a good time to refresh your memory on integration techniques from Calculus 1 and 2. It is recommended that you review sections 5.5 (integration by substitution) and 7.1 (integration by parts) especially.

Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

- **Type I region:** [Discussion above Figure # 5 in Section 15.2.]

  a region that lies between two continuous functions of x

  $$D = \left\{(x,y) \in \mathbb{R}^2 \,/\, x \in [a,b],\ y \in [g_1(x), g_2(x)]\right\}$$

- **Type II region:** [Discussion above Figure #7 in Section 15.2.]

  a region that lies between two continuous functions of y

  $$D = \left\{(x,y) \in \mathbb{R}^2 \,/\, y \in [c,d],\ x \in [h_1(y), h_2(y)]\right\}$$

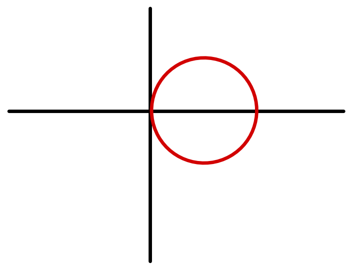
Topics to learn in this section:

1. Integration over a general region.
2. The geometric meaning of Fubini's Theorem: slicing the area in two different ways.
3. Properties of double integrals.

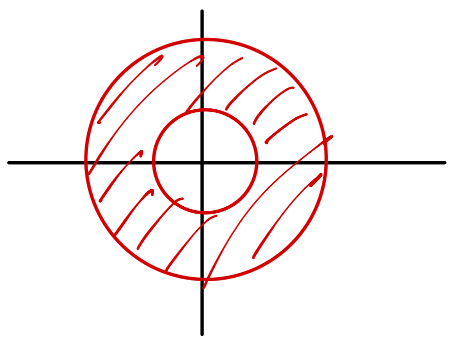
15.3 Double Integrals in Polar Coordinates

Prerequisite information for this section:

1. Sketch the region in the (polar) plane described by the polar coordinate equation, $r = \cos 2\theta$, where $0 \le \theta \le 2\pi$. (You may wish to review material from Calculus 2, section 10.3 and 10.4 of your text. You may also wish to use your graphing calculator, set to polar coordinates.)



2. Sketch the region in the (polar) plane described by the inequality, $2 \le r \le 5$, for $0 \le \theta \le 2\pi$. Use your knowledge of geometry to find the area of this region, assuming the unit of measure is a centimeter.



Definitions in this section: Give the definition of each term below, using the given passage from your text/e-text.

- **Polar rectangle:** [Above Figure 3 in Section 15.3.]

A set of the form $R = \{(r,\theta) \mid r \in [a,b],\ \theta \in [\alpha,\beta]\} \subset \mathbb{R}^2$

Topics to learn in this section:

1. The definition of a polar rectangle: what it looks like, and its differential area $r dr d\theta$.
2. The idea that some integrals are simpler to compute in polar coordinates.
3. Integration over general polar regions.